

Continuity Equation Derivation for Square Contraction Nozzle

$$Q = A_1 \cdot V_1 = A_2 \cdot V_2 \quad (1)$$

Where:

- A_1, A_2 = Cross-sectional areas of the inlet and outlet (m^2)
- V_1, V_2 = Air velocities at the inlet and outlet (m/s)

1. Geometric Area Calculations

Inlet side length is 210 mm (0.21 m) and the outlet side length is 78 mm (0.078 m):

$$A_1 = 0.21 \text{ m} \times 0.21 \text{ m} = 0.0441 \text{ m}^2 \quad (2)$$

$$A_2 = 0.078 \text{ m} \times 0.078 \text{ m} = 0.006084 \text{ m}^2 \quad (3)$$

2. Determining the Contraction Ratio

The geometric area ratio is determined by dividing the inlet area by the outlet area:

$$\text{Contraction Ratio} = \frac{A_1}{A_2} = \frac{0.0441 \text{ m}^2}{0.006084 \text{ m}^2} \approx 7.25 \quad (4)$$

The exit velocity relationship simplifies directly to:

$$V_2 = 7.25 \cdot V_1 \quad (5)$$

3. Velocity Solution from Volumetric Flow Rate (Q)

With a measured flow rate of 54.33 CFM, we convert imperial units to SI metric units ($1 \text{ CFM} = 0.00047195 \text{ m}^3/\text{s}$):

$$Q = 54.33 \times 0.00047195 = 0.02564 \text{ m}^3/\text{s} \quad (6)$$

By rearranging Equation (1) to solve for velocity ($V = \frac{Q}{A}$), we find the localized airspeeds:

$$V_{\text{inlet}} = \frac{0.02564 \text{ m}^3/\text{s}}{0.0441 \text{ m}^2} \approx 0.58 \text{ m/s} \quad (7)$$

$$V_{\text{outlet}} = \frac{0.02564 \text{ m}^3/\text{s}}{0.006084 \text{ m}^2} \approx 4.21 \text{ m/s} \quad (8)$$

4. Total Velocity Gain

The net differential velocity (ΔV) generated exclusively by the nozzle architecture yields:

$$\Delta V = V_{\text{outlet}} - V_{\text{inlet}} = 4.21 \text{ m/s} - 0.58 \text{ m/s} = 3.63 \text{ m/s} \quad (9)$$